

EXPERIMENT - 1

Linux Programming Environment & Basic Commands

I.Objective

To understand and set up a Linux programming environment using a Virtual Machine Manager and to execute basic Linux commands.

2. Setting up of Linux Programming Environment

C programming can be done online or offline according to the user's needs. For our requirements, it is essential to work within a Linux Environment.

2.1 Online Environment

Cocalc.ai: Cocalc is an online collaborative workspace that provides instant access to Linux environments.

The steps followed to use it include:

1. Navigate to Cocalc.com and create a free account or log in.
2. Create a new "Project" for the programming tasks.
3. Within the project, open a new "Linux Terminal" or a Jupyter Notebook with a C kernel.
4. Write the C code using editors like nano or vim, compile it using the gcc compiler, and execute the generated binary directly in the browser terminal.

2.2 Offline Environments

Offline setups run locally on the user's hardware. Common methods include:

- Dual Boot: Installing two operating systems (e.g., Windows and Linux) on the same hard drive. The user selects which OS to load at startup.
 - Single Boot: Wiping the existing operating system and installing only Linux on the machine, dedicating all hardware resources to it.
 - Bootable Pendrive (Live USB): Flashing a Linux ISO onto a USB drive and booting directly from it without modifying the host hard drive. Changes may or may not be persistent
 - Virtual Machine (VM): Running Linux as a guest operating system inside a software emulator (Hypervisor) on top of the host OS. This is the method chosen for this practical.
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3. Virtual Machine Manager (Hypervisor)

A hypervisor is software that creates and runs virtual machines, allowing one host computer to support multiple guest VMs by virtually sharing its resources.

3.1 Type-1 (Baremetal)

A bare-metal hypervisor installs directly onto the computing hardware, acting as the host operating system itself. It is highly efficient and typically used in enterprise servers.

Examples: VMware ESXi, Microsoft Hyper-V, Citrix Hypervisor (XenServer).

3.2 Type-2 (Hosted)

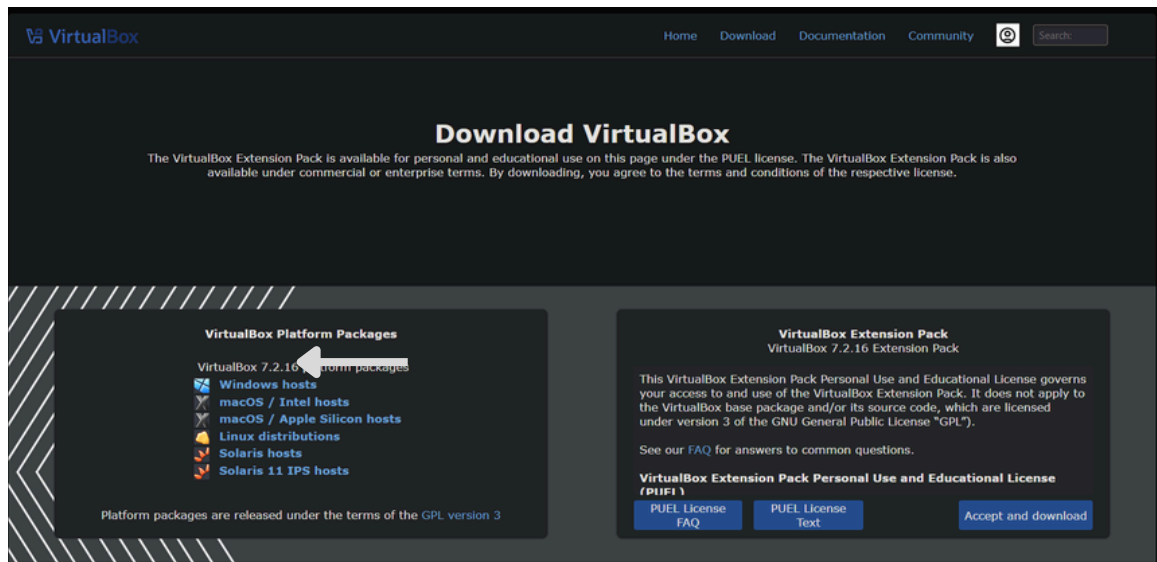
A hosted hypervisor runs as a software layer or application on top of a conventional operating system (like Windows or macOS).

Examples: Oracle VirtualBox, VMware Workstation, Parallels Desktop

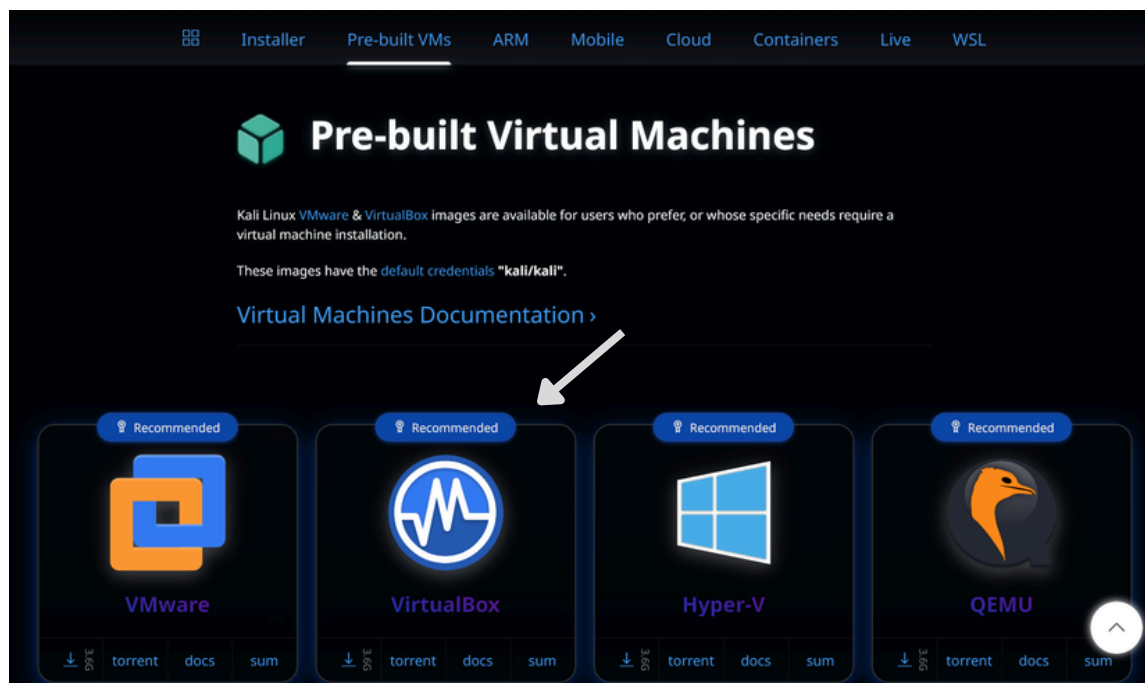
4. Installation of VMM & VM (Method Used)

For this experiment, Oracle VirtualBox (Type-2 Hypervisor) and a pre-built Kali Linux VM were utilized. The following steps were executed:

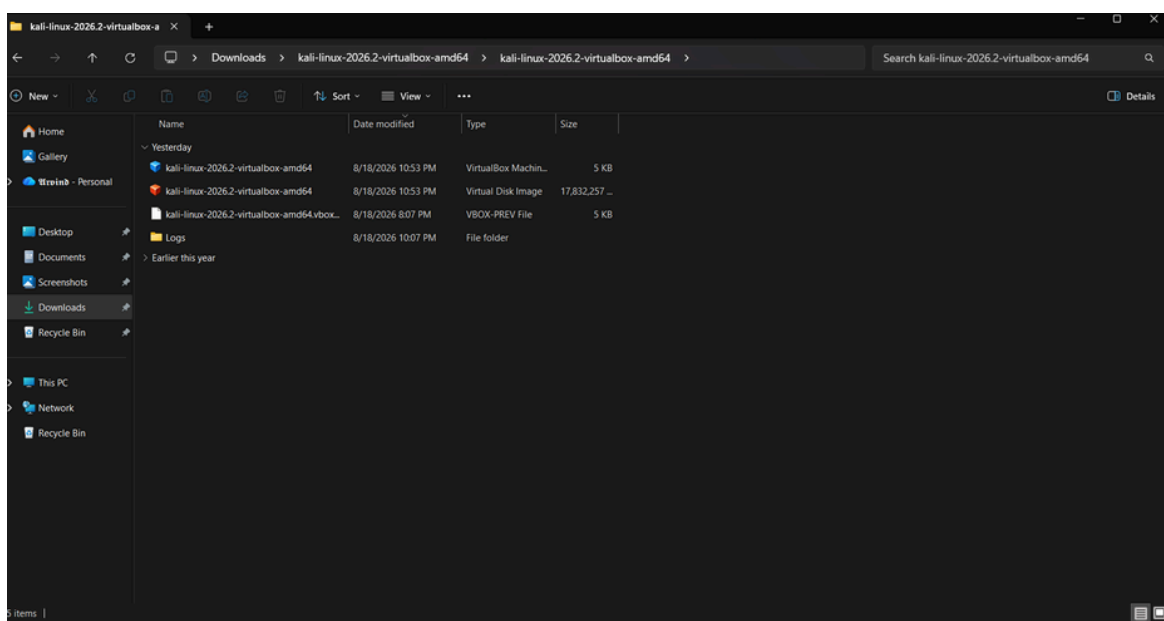
- Step 1: Downloaded the Oracle VirtualBox installer for "Windows hosts" from the official website.



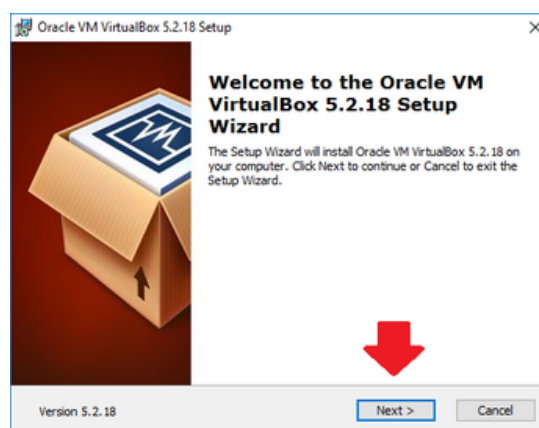
- Step 2: Downloaded the Pre-built Kali Linux VM archive specifically configured for VirtualBox from the Kali Linux official site.



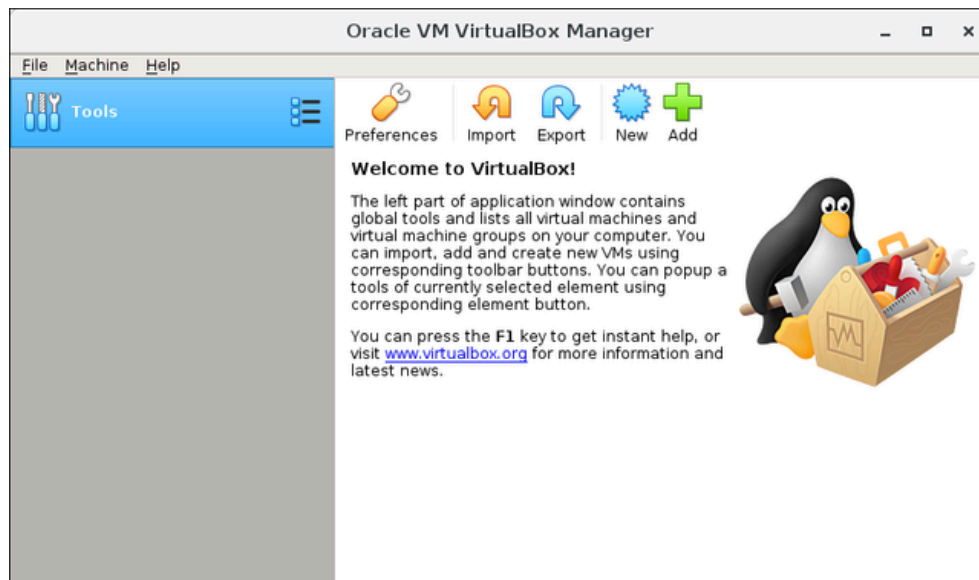
- Step 3: Extracted the downloaded Kali Linux `.7z` file to a local directory.



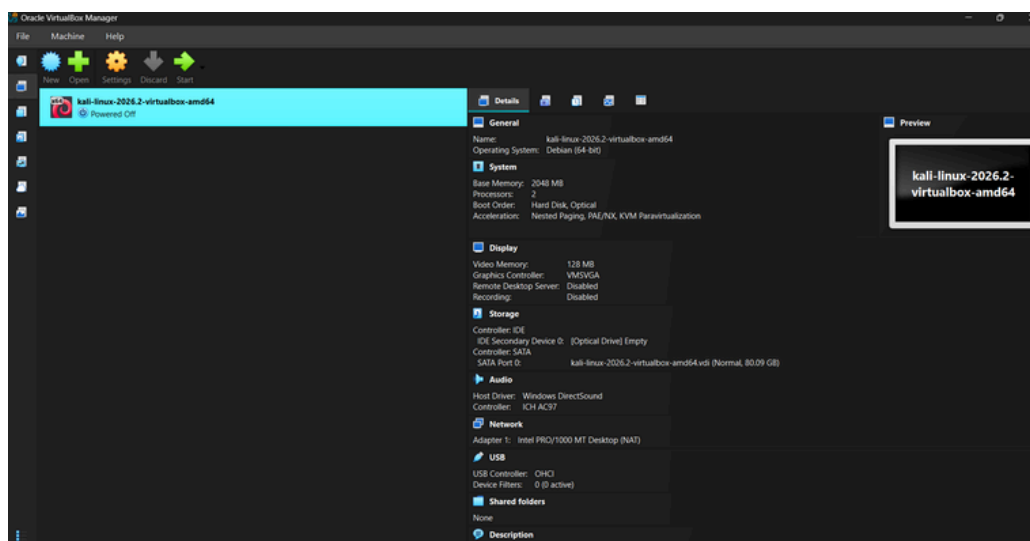
- Step 4: Installed VirtualBox on the host Windows machine by running the executable and following the on-screen instructions.



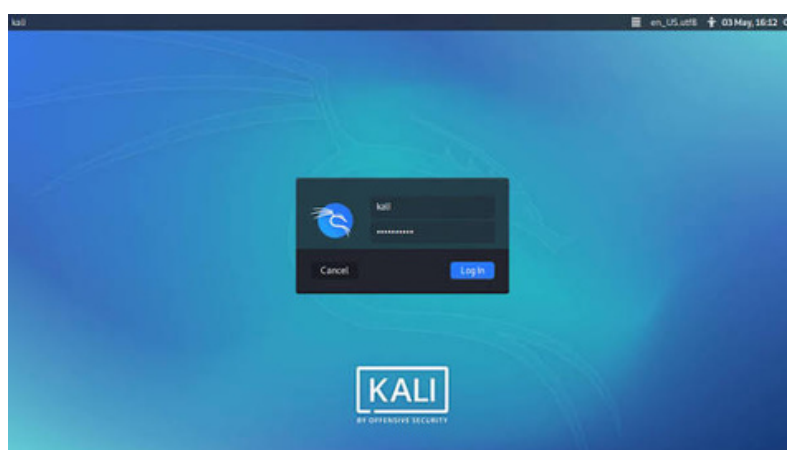
- Step 5: Open the Oracle VirtualBox Manager.



- Step 6: Opened the "saved VM" by navigating to Machine > Add and selecting the extracted .vbox file of Kali Linux and Start the Virtual Machine



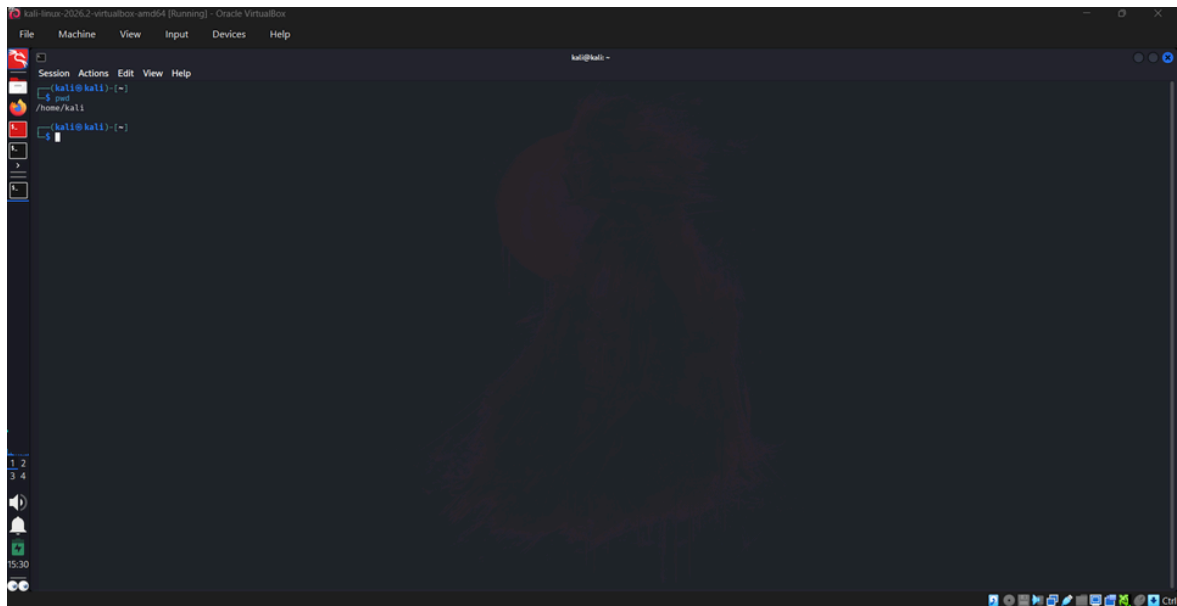
- Step 7: Boot the virtual machine, logging in using the default credentials (kali/kali).



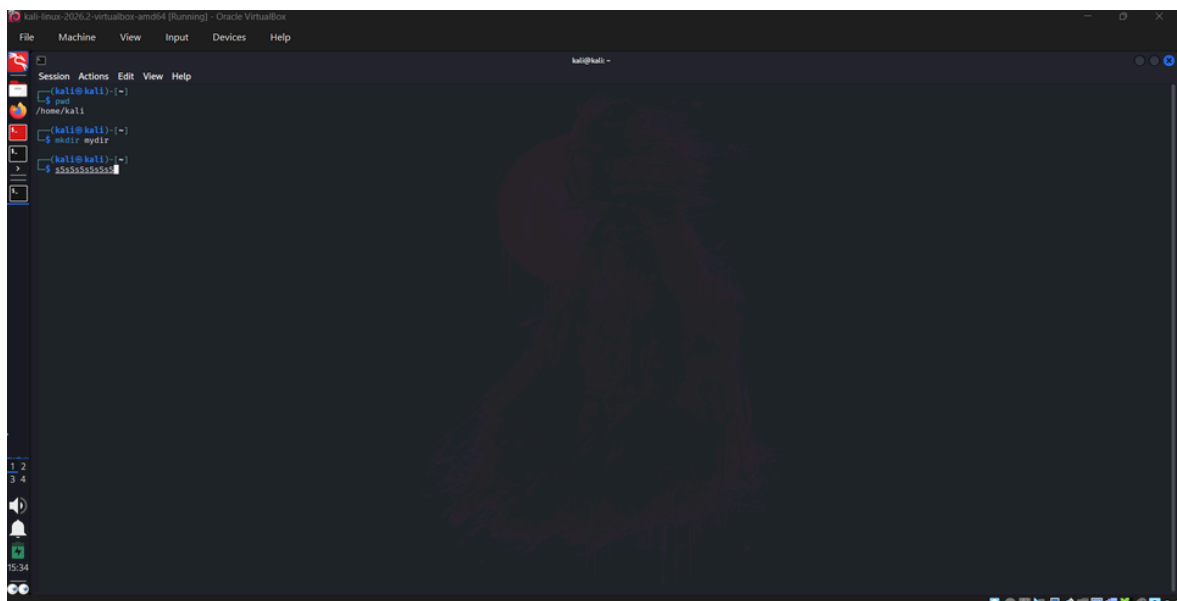
Conclusion: The installation of the VMM (Oracle VirtualBox) and VM (Kali Linux) was successfully completed.

5. Working with Linux Commands

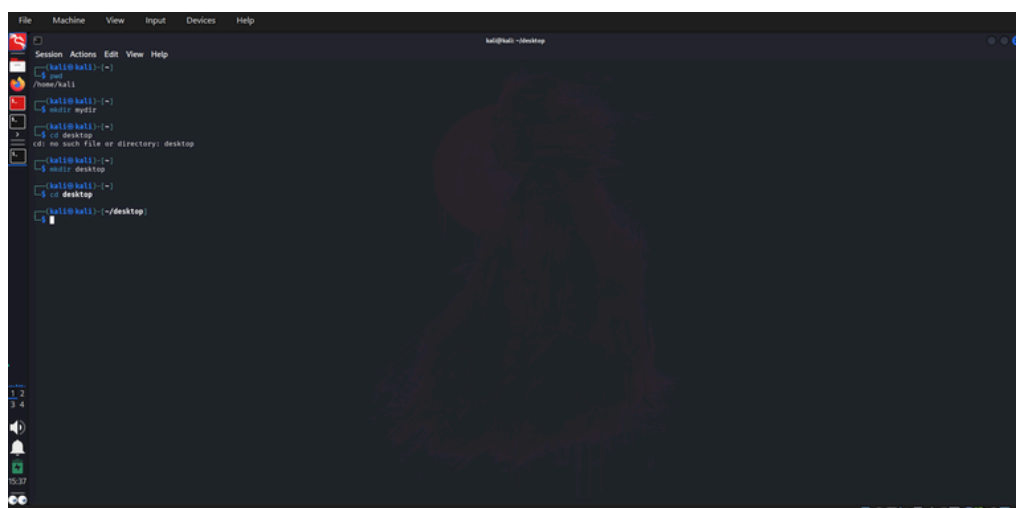
1. pwd: Prints the absolute path of the current working directory.



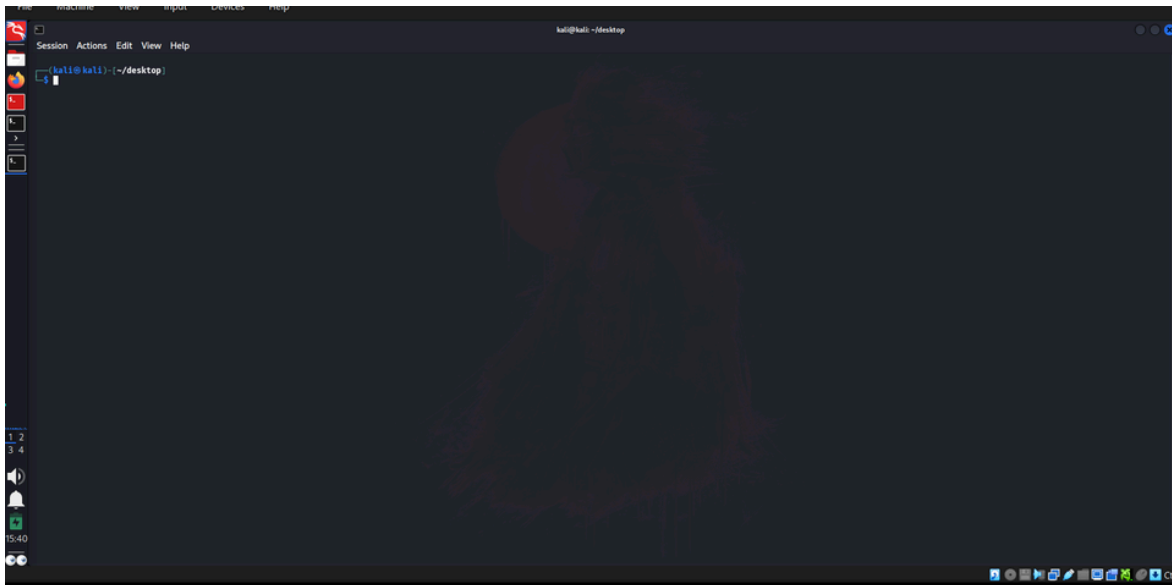
2. mkdir mydir Creates a new directory named 'mydir' in the current path



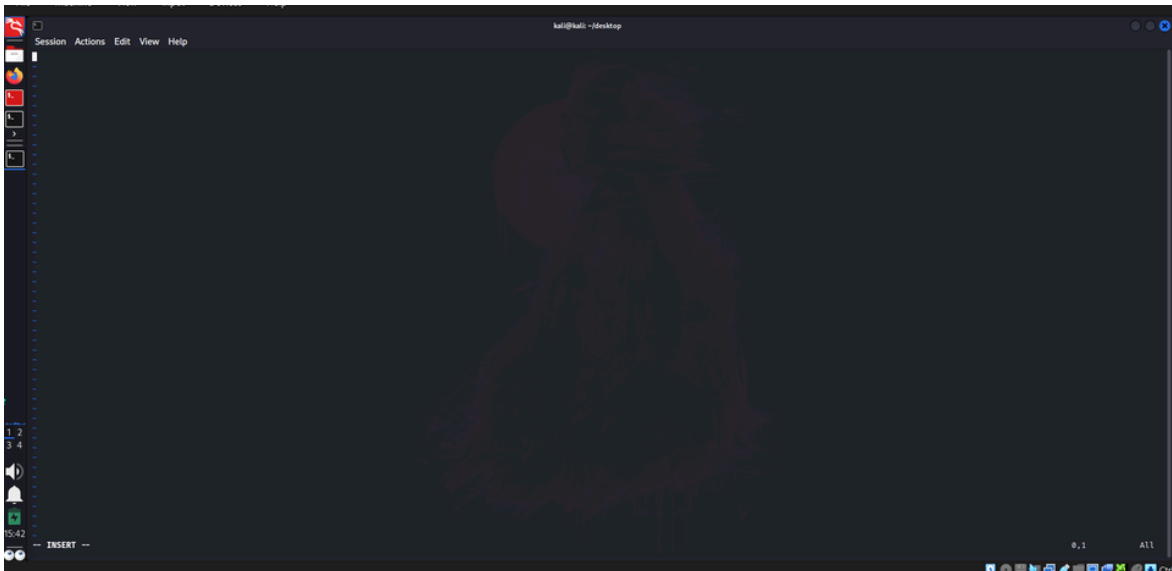
3. cd desktop: Changes the current directory to 'Desktop'.



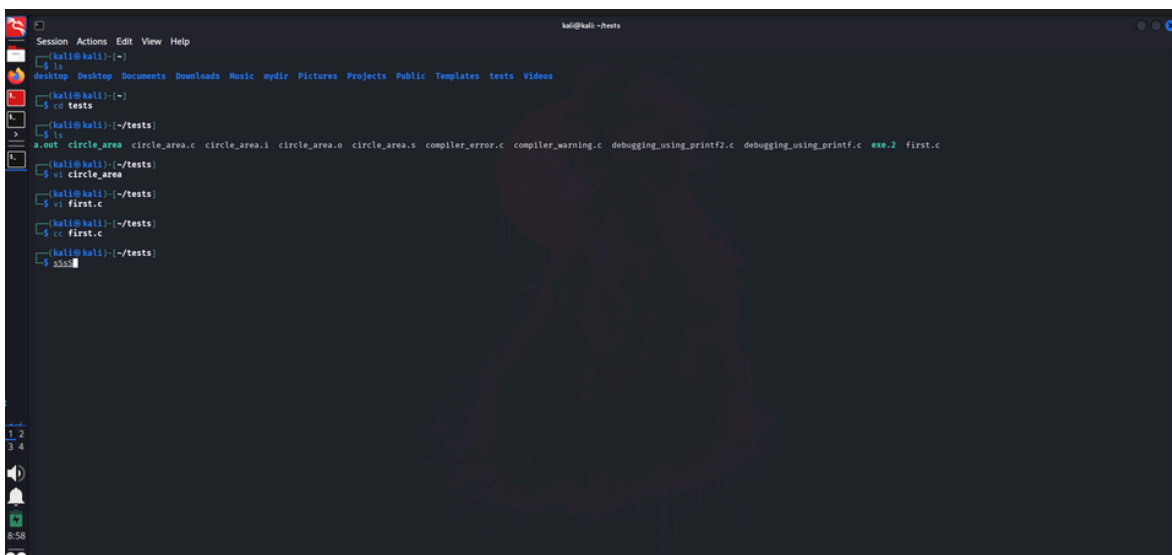
4. Clear: Clears the output of the terminal.



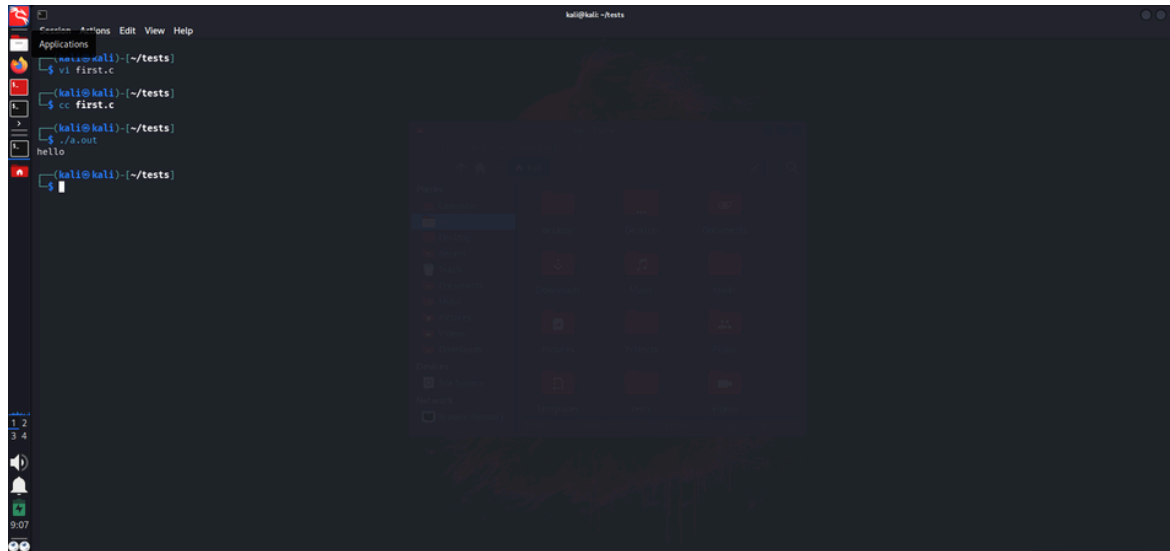
5. vi first.c: Open the editor for the new created file of name first by pressing i we enter into insert mode



6. cc first.c: A compiler command to compile the program.



7. `./a.out` : After the compilation a executable file is created by default it is `a.out`, to have a output we need to put this command

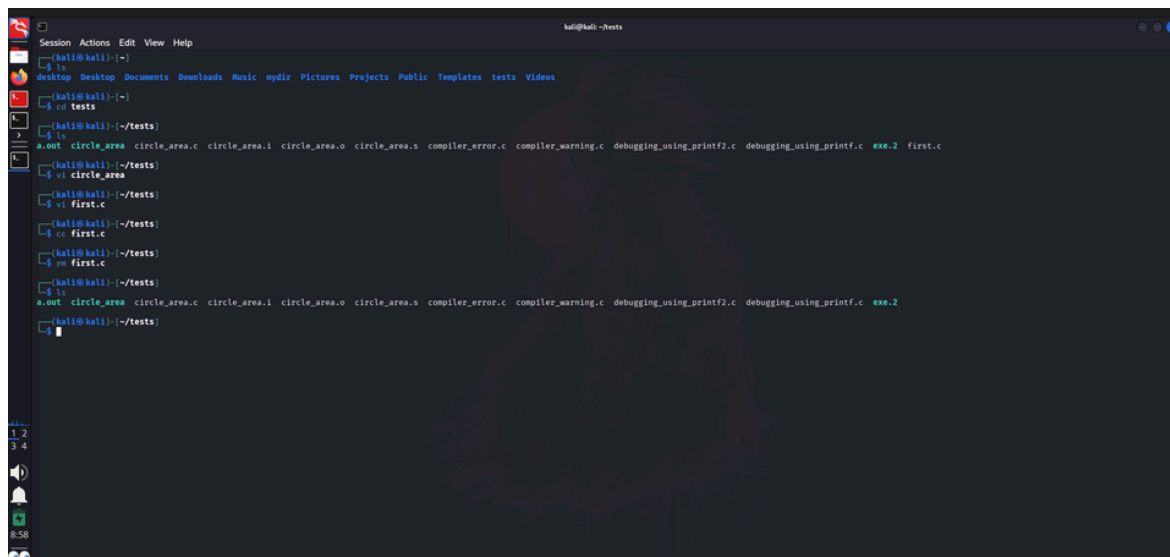


A screenshot of a terminal window with a dark background. The terminal shows the following commands and output:

```
kali@kali:~/tests
$ vi first.c
$ cc first.c
$ ./a.out
hello
```

The terminal window has a menu bar with 'Session', 'Actions', 'Edit', 'View', and 'Help'. On the left side, there is a sidebar with icons for 'Applications', 'Desktop', 'Documents', 'Downloads', 'Music', 'mydir', 'Pictures', 'Projects', 'Public', 'Templates', 'tests', and 'Videos'. The bottom status bar shows the time as 9:07.

7. `rm first.c` : Remove the file or directory



A screenshot of a terminal window with a dark background. The terminal shows the following commands and output:

```
kali@kali:~/tests
$ ls
$ cd tests
$ ls
a.out circle_area circle_area.c circle_area.i circle_area.o circle_area.s compiler_error.c compiler_warning.c debugging_using_printf2.c debugging_using_printf.c exe.2 first.c
$ vi circle_area
$ vi first.c
$ cc first.c
$ rm first.c
$ ls
a.out circle_area circle_area.c circle_area.i circle_area.o circle_area.s compiler_error.c compiler_warning.c debugging_using_printf2.c debugging_using_printf.c exe.2
```

The terminal window has a menu bar with 'Session', 'Actions', 'Edit', 'View', and 'Help'. On the left side, there is a sidebar with icons for 'Applications', 'Desktop', 'Documents', 'Downloads', 'Music', 'mydir', 'Pictures', 'Projects', 'Public', 'Templates', 'tests', and 'Videos'. The bottom status bar shows the time as 9:58.